MODULE 3 – exercise :3

Exercise 5: Return Data from a Stored Procedure

Aim

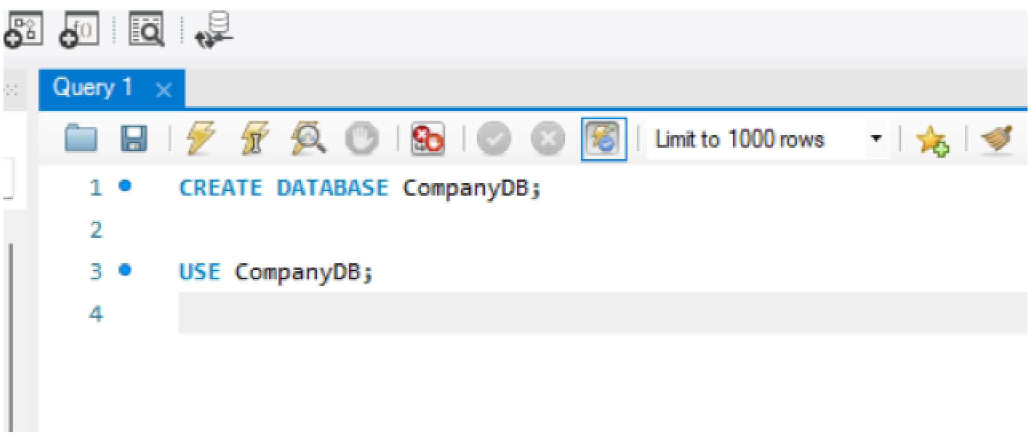
To create a stored procedure that returns the total number of employees working in a specific department.

Theory

A **Stored Procedure** is a collection of SQL statements that are stored permanently in the database. It can accept input parameters, perform operations, and return results. Stored procedures improve code reusability, security, and performance.

In this exercise, a stored procedure is created that accepts a **DepartmentID** as an input parameter and returns the total number of employees in that department using the **COUNT()** aggregate function.

Create Database :



Output:

Create Employee Table :

```
CREATE TABLE Employees
(
    EmployeeID INT PRIMARY KEY,
    EmployeeName VARCHAR(50),
    DepartmentID INT,
    Salary DECIMAL(10,2)
);
show tables;
```

Output :

✔	6	07:54:16	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
✔	7	07:54:57	CREATE TABLE Employees (EmployeeID INT PRIMARY KEY, EmployeeName VARCHAR(50), Depart...	0 row(s) affected

Insert Data:

```
INSERT INTO Employees
VALUES
(101,'Rahul',1,50000),
(102,'Priya',1,55000),
(103,'Kiran',2,60000),
(104,'Sneha',2,58000),
(105,'Arjun',1,62000),
(106,'Ravi',3,45000),
(107,'Meena',3,47000);
SELECT * FROM Employees;
```

Output :

Result Grid | Filter Rows: | Edit: | Export/

EmployeeID	EmployeeName	DepartmentID	Salary
101	Rahul	1	50000.00
102	Priya	1	55000.00
103	Kiran	2	60000.00
104	Sneha	2	58000.00
105	Arjun	1	62000.00
106	Ravi	3	45000.00
107	Meena	3	47000.00
NULL	NULL	NULL	NULL

Create Stored Procedure:

```sql
DELIMITER $$

CREATE PROCEDURE GetEmployeeCount(IN DeptID INT)
BEGIN
    SELECT COUNT(*) AS TotalEmployees
    FROM Employees
    WHERE DepartmentID = DeptID;
END $$

DELIMITER ;
CALL GetEmployeeCount(1);
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TotalEmployees
3

For Department 2:

```sql
CALL GetEmployeeCount(2);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TotalEmployees
2

For Department 2:

```sql
CALL GetEmployeeCount(3);
```

TotalEmployees
2

Verify the Data :

```sql
SELECT * FROM Employees;
```

EmployeeID	EmployeeName	DepartmentID	Salary
101	Rahul	1	50000.00
102	Priya	1	55000.00
103	Kiran	2	60000.00
104	Sneha	2	58000.00
105	Arjun	1	62000.00
106	Ravi	3	45000.00
107	Meena	3	47000.00
NULL	NULL	NULL	NULL

Employees 6